Matteo Jiahao Chen

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Education

Tsinghua University, Beijing Sep 2024 – Jun 2027 (expected)
M.Sc. Computer Science; GPA: 3.8/4; full scholarship; advisors: Yaqin Zhang, Yuanchun Li.

Politecnico di Torino, Turin Sep 2020 – Jul 2024
B.Eng. Computer Engineering; final grade: 100/110; full scholarship.

Tongji University, Shanghai Information Technology exchange Sep 2021 – Sep 2022

Work Experience

Institute for AI Industry Research (AIR), Tsinghua University Feb 2026 – Present
Research Intern, Beijing; ongoing SSPC and Duplex research.

- Developing **Sparse Semantic Point Cloud (SSPC)**, a task-adaptive 3D memory for robotic manipulation linking point coordinates, object descriptions and temporal histories, with denser sampling in task-relevant regions.
- Implemented semantic point generation, task-dependent sampling and a training-data curation pipeline pairing task instructions and full videos with semantic 3D point targets; data curation is ongoing, with evaluation pending.
- Benchmarked vision in a **Qwen3.5**-based real-time speech/text/video interaction model (**Duplex**): interactive visual understanding, temporal recall, FPS, latency and streaming performance.
- Prepared video-question-answer instruction-tuning data for interactive video understanding.

Research

Feedback-Driven Test-Time Scaling for Automated Theorem Proving Under review
Tsinghua Knowledge Engineering Lab collaboration with Zhipu.

- Built an adaptive proof-data pipeline: generated harder/easier problems based on Lean solvability, then formalized and solved them with Kimina-Autoformalizer and **DeepSeek-Prover-V2-671B**.
- Post-trained **Qwen3-8B** with SFT, GRPO and expert iteration on approximately **40K** RL examples; used Lean compiler feedback and `extract_goal` to repair failed proofs and recycle them into training.
- Evaluated Lean-feedback MCTS separately on **DeepSeek-Prover-V2-7B**: MiniF2F-Test **73.77%** vs. **69.67%** (**+4.10 pp**) and PutnamBench **42** vs. **33** solved, both against standard MCTS at Pass@12.

TimeColor: Flexible Reference Colorization via Temporal Concatenation ICME 2026; fourth author

- Helped build a **CogVideoX-5B** model that colors sketch videos from arbitrary-frame, character and background references; contributed to data preparation (**120K** single-/**96K** multi-reference examples), implementation, training and evaluation.
- Helped implement temporal latent concatenation, modality-disjoint RoPE and reference-region attention masks to limit color leakage. Team FVD on Sakuga-42M: **30.74** vs. **63.44** (AniDoc, first-frame); **47.13** vs. **157.35** (ToonComposer, multi-reference).

Projects

Framework for Seamless Advertising in LLMs github.com/HectorRguez/Ads

- Built a Flask/vLLM system to insert relevant ads into LLM answers: DeepSeek V3 generates the answer, Stella-en-1.5B retrieves an advertiser, and **Mistral-7B-Instruct** rewrites the response to incorporate the ad.
- Fine-tuned with LoRA + DPO on **641** human-labeled preferences from **2,998** WildChat pairs; TRUE factual consistency rose from **0.731** to **0.735** (**0.739** without ads) across 10 datasets of 1,000 examples each.

Financial Information Agent

- Built a **DeepSeek-R1-Distill-Qwen-32B** agent and runtime harness with FAISS RAG over company reports/PDFs and schema-aware SQL; added SQL validation, error-driven retries and evidence checks when needed.
- Achieved **95.7%** final-answer accuracy vs. **78.5%** for simple RAG (**+17.2 pp**) on a **350**-question internal benchmark, using the same questions and predefined reference answers.

Chessformer: Check Your Attention github.com/Nech07/searchless_chess

- Trained a Transformer to choose chess moves without tree search using existing **ChessBench** data (10M games; 15.3B Stockfish 16-labeled pairs); ran attention and positional-encoding ablations in JAX.
- Smolgen raised puzzle accuracy from **80.41%** to **85.30%** and puzzle-estimated Elo from **2018** to **2148**; bilinear attention improved estimated Elo by **49** at the same parameter count.

Awards

ICPC/SWERC 2022–23: **37/120**; ITACPC 2022: **5th in Italy**; IBEC 2023: **1/107**.
Tianchi Repeat Buyers 2024: **7/39,037**; CyberChallenge 2022 (pwnthem0le): CTFtime global rank **23/38,050**.

Skills

Languages: Italian (native), English (IELTS 7.5), Chinese (HSK 5).
Technical: Python, C/C++, Linux, PyTorch, Transformers, DeepSpeed, LlamaFactory, TRL, vERL, vLLM, SGLang, llama.cpp, Lean 4.